

First-Update or Last-Update?

Position-dependent retrieval when the same key is updated many times in context — what the Qwen + Gemma families tell us so far.

THE PROBLEM & RESEARCH GOAL

Does the position of information in context change retrieval?

ICU monitor → log file → LLM directly answering queries

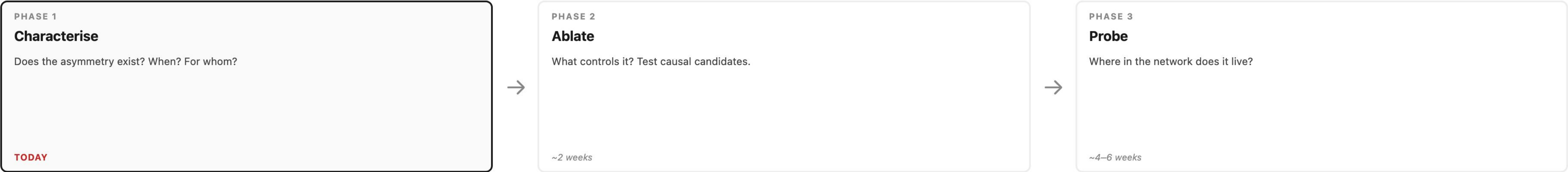
[14:02] HR=82 BP=128/82 SpO2=98 Temp=37.1
[14:17] HR=88 BP=125/80 SpO2=97 Temp=37.3
[14:32] HR=94 BP=118/76 SpO2=96 Temp=37.6
[14:47] HR=102 BP=110/72 SpO2=94 Temp=38.0
[15:02] HR=115 BP=98/64 SpO2=92 Temp=38.4
[15:17] HR=128 BP=92/58 SpO2=90 Temp=38.7

Query	Expected	What it tests
"What was the heart rate at admission?"	82	Recall first value
"What is the patient's current heart rate?"	128	Recall most-recent value

Why this isn't Lost-in-the-Middle

	Lost in the Middle	This work
Information structure	One fact + many <i>static distractors</i>	Same key, repeatedly overwritten
Context fill	Near-saturated (often >75%)	Moderate (5–30%)
Failure type	Retrieval of unique target	Conflict between valid alternatives
Question	"Where does the model look?"	"Which version wins?"

RESEARCH GOAL *three-phase arc — today is Phase 1 only*



Today's discussion: Phase 1 results, open questions, planned ablations. Want feedback on (a) **task-design cleanliness**, (b) **which open question to chase first**, (c) **whether the planned controls are right**.

TASK DESIGN & DATASETS

Controlled key–value interference task across three datasets

Stream of K keys, each updated N times in random order:

Key1: value_a
Key2: value_b
Key1: value_c ← Key1 updated
Key3: value_d
Key1: value_e ← Key1 updated again
...

RI · RETROACTIVE
"What was the *first* value of Key1?"
expected: value_a

PI · PROACTIVE
"What is the *current* value of Key1?"
expected: value_e

INDEPENDENT VARIABLES

K · KEYS {2,3,5,7,10,15,20,25,30}	N · UPDATES {5,7,10,15,20,30,50,75,100}	TRIALS 100/cell	GRID 9 × 9
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K = working-memory load; N = interference depth. Sweeping both separates the two — most prior work confounds them.

· Greedy decoding (T=0) · Exact-match · Wilson 95% CIs · ~16k trials/dataset/model

Methodology note: first-sweep garbage rate **60–80%** → tightened prompt → **<6%**. All today's numbers post-fix.

THREE DATASETS design as falsification — each removes one suspected confound

SINGLE-TOKEN, NO PRIORS

ARBITRARY-SINGLE

2,300 single-token English words across 46 categories. The key is the word itself, repeated at new context positions.

Removes confound: Forces purely positional retrieval (no semantic priors).

SAMPLE STREAM

apple ← first
mountain
iron
apple
steel
ocean
...
apple ← last

RI first 'apple'? → position 1

PI last 'apple'? → position 7

MULTI-TOKEN, REAL CATEGORIES

SEMANTIC-MULTI

2,403 multi-token values anchored to real categories (fruit, metal, country...). Per-category pools enforce within-category plausibility.

Removes confound: Tests vocabulary realism and tokenization.

SAMPLE STREAM

fruit: blueberry ← first
metal: copper
country: Portugal
fruit: papaya
city: Bangkok
metal: titanium
...
fruit: dragon fruit ← last

RI first 'fruit'? → blueberry

PI last 'fruit'? → dragon fruit

NATURALISTIC PROSE

NARRATIVE

Generated Dota-2 commentary prose with implicit hero-pick updates embedded in story flow — not an artificial KV format.

Removes confound: Tests real-world prose, not artificial KV format.

SAMPLE STREAM

Player1 picks Pudge. ← first
Player2 picks Anti-Mage.
Player1 swaps to Sniper.
Player1 swaps to Lina.
Player3 swaps to Lion.
...
Player1 swaps to Zeus. ← last

RI Player1's first pick? → Pudge

PI Player1's current pick? → Zeus

Qwen + Gemma — First-Update vs Last-Update Retrieval

Full-grid means · 100 trials/cell · two open-weight families compared at matching scales

ARBITRARY-SINGLE

single-token, no semantic priors

	RI recall first	PI recall last	Gap RI – PI	cells
Qwen 0.5B Inst	40%	9%	+31.0%	n=78
Qwen 1.5B Inst	63%	16%	+46.4%	n=79
Qwen 3B Inst	53%	50%	+2.5%	n=79
Qwen 3B Base <i>(no SFT)</i>	44%	56%	-11.8%	n=73
Gemma 270M It	18%	3%	+15.3%	n=63
Gemma 1B It	57%	3%	+54.3%	n=55
Gemma 4B It	95%	49%	+45.9%	n=76

SEMANTIC-MULTI

multi-token, real categories

	RI recall first	PI recall last	Gap RI – PI	cells
Qwen 0.5B Inst	57%	17%	+39.8%	n=63
Qwen 1.5B Inst	60%	28%	+32.4%	n=63
Qwen 3B Inst	63%	58%	+4.8%	n=63
Qwen 3B Base <i>(no SFT)</i>	50%	65%	-15.1%	n=61
Gemma 270M It	13%	6%	+6.5%	n=51
Gemma 1B It	57%	6%	+51.1%	n=45
Gemma 4B It	94%	63%	+31.9%	n=63

RI high (good) PI low (bad) Gap > 0 — primacy bias (RI > PI) Gap < 0 — recency wins (PI > RI)

RI = recall first value · PI = recall last value · Gap = RI – PI · n_cells = (K, N) cells averaged

THREE REGIMES — AND A CROSS-FAMILY PUZZLE

STRONG PRIMACY BIAS

Qwen 0.5B (+31%) · 1.5B (+46%)
Gemma 1B (+54%) · 4B (+46%)

"Default for instruct models"

SYMMETRIC

Qwen 3B Inst (+2.5%)

only model in our sample that escapes

"Capacity isn't the explanation —
Gemma 4B doesn't escape"

REVERSED

Qwen 3B Base (–12%)

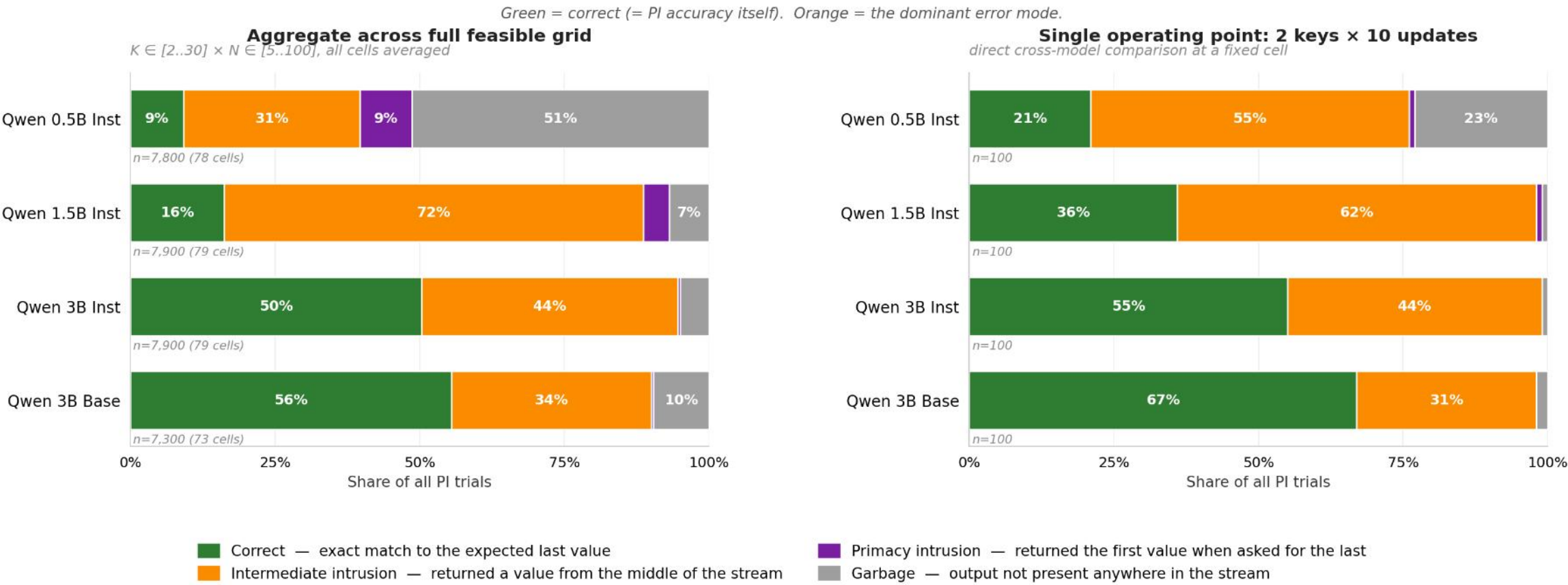
same architecture, no SFT

"Removing SFT flips it"

The asymmetry is not universal — and not monotone with scale. Gemma 4B (4× the parameters of Qwen 1.5B) still shows +46% gap. Variation is the finding.

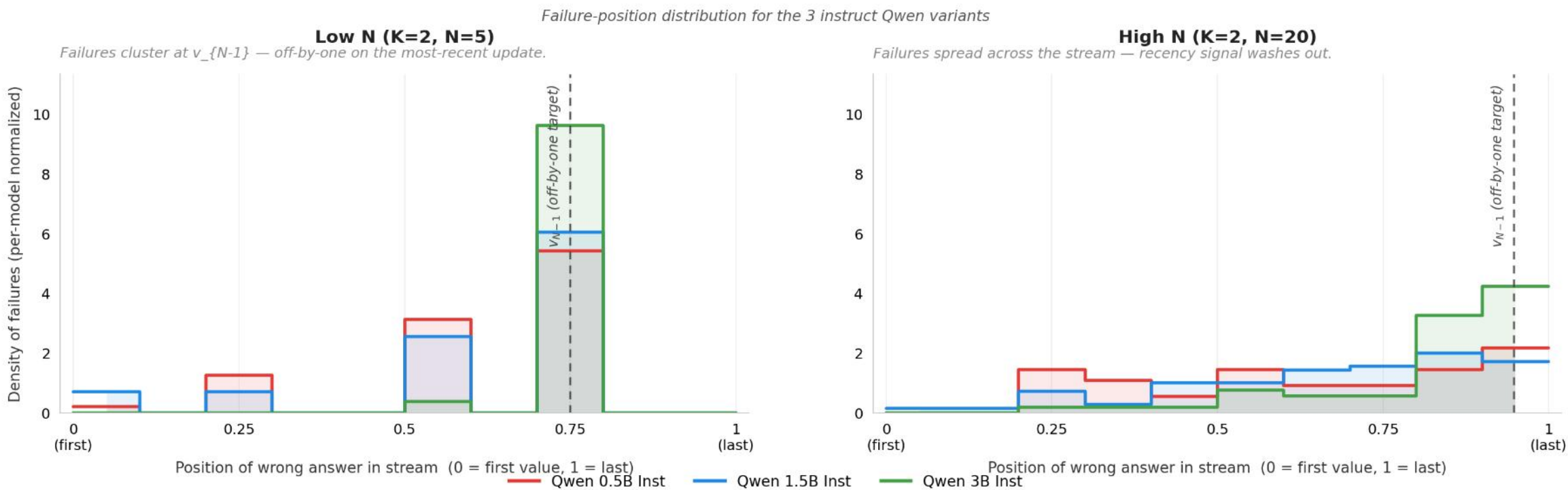
ERROR CLASSIFICATION **How the model fails — not just whether.**

How does PI fail? — Error-type breakdown across the Qwen family



Dataset: ARBITRARY-SINGLE · 100 trials per cell · greedy decoding · exact-match scoring

Where do PI failures land? — Off-by-one at low N, diffuse at high N



Source: ARBITRARY-SINGLE · 100 trials per cell

TAKEAWAY · At **low N**, PI failures cluster off-by-one (penultimate value). At **high N**, they spread diffusely. **Failure mode is N-dependent** — motivates a Lost-in-the-Middle-style scan over update positions as the next experiment.

OPEN QUESTIONS & NEXT EXPERIMENTS

Each open question has a planned ablation that answers it.

Phase 2 (rows 1–8) ~1–2 weeks each; Phase 3 (row 9) only after Phase 2 narrows the candidate causes.

Q	OPEN QUESTION	PLANNED NEXT EXPERIMENT	WHAT IT TELLS US
1	Why does Qwen 3B Instruct escape the asymmetry while Gemma 4B does not?	→ Capacity vs training-data ablation	Match parameter count across families and vary instruction-tuning data; isolate which factor breaks primacy bias.
2	Why does removing SFT flip the direction (recency bias instead of primacy)?	→ SmoLLM3 checkpoint sweep	Run the task across pretraining → SFT → alignment checkpoints to see when in training the asymmetry emerges and which stage flips it.
3	Is the failure surface U-shaped (LITM), single-sided (recency-collapse), or architecture-specific?	→ Per-position accuracy(i) scan + Liu et al. replication	Query every update index $i \in [0, N-1]$, plot accuracy(i). Also run Liu et al.'s exact prompt as a positive control — if their U-shape reproduces on our models, our finding is task-specific, not model-specific.
4	Is the asymmetry interference-driven, or just positional-decay-driven?	→ Interleaved vs blocked update ordering	Re-run the same K, N grid with all updates for a key contiguous (blocked) instead of randomly interleaved. If the gap survives blocking, it's positional decay (LITM-flavored). If the gap dies, we've earned the word 'interference'.
5	Does LITM hold under varying context fill, or is fill orthogonal to the gap?	→ Saturation sweep	Vary context fill (5%, 25%, 75%, 95%) at fixed N via neutral padding. LITM is worst at 75–95% fill; if our gap is flat across fill, LITM is actively rejected — not just untested.
6	Does the effect depend on positional addressing or content binding?	→ Numbered-update probe	Tag every update with an explicit integer (Update 7: ...) and query by tag. If the gap collapses, the bias is positional addressing, not content binding.
7	Is the asymmetry chat-template-induced, or model-intrinsic?	→ Multi-turn vs single-turn re-formatting	Re-encode the same K, N stream as N separate user turns (ChatML role tags between updates). If the gap shape changes, the chat template — not the model — is producing it. Could explain the SFT-flip.
8	Can the model retrieve first values when shown how, or is it incapable?	→ Few-shot demonstrations	Add 2–3 in-context examples of correctly answering 'first' queries before the test query. If the gap collapses, the bias is a default — not a capacity limit. Sharpens 'models can't' into 'models won't'.
9	Where in the network does the surviving mechanism live?	→ Phase 3 — mechanistic battery	After Phase 2 narrows candidates: logit lens (when v_first vs v_last appear in the residual stream), attention-pattern decomposition, activation patching, probing classifiers. Partly piloted on SmoLLM2 already.